

Security in LoRaWAN networks



LoRa (Long Range)

- ▷ Low-power, long-distance data wireless communication technology
 - ♦ Radio technology
- ▷ Mainly used in Internet of Things (IoT) applications
 - ♦ Devices need to send small amounts of data over several kilometers while consuming very little energy

LoRaWAN (LoRa Wide Area Network)

- ▷ Network protocol on top of LoRa
 - ◆ Defines rules and architecture for how devices communicate within a network
 - ◆ It manages aspects like
 - Device authentication and security
 - Message routing through Gateways and Network Servers
 - Data delivery between End devices and Application Servers
 - Network scalability and management
- ▷ LoRaWAN supports secure, bi-directional communication between IoT devices and central servers
 - ◆ Ideal for smart cities, agriculture, environmental monitoring, and industrial automation

LoRa/LoRaWAN technical features

▷ Low data rate

- ♦ Optimized for small data packets
 - A few bytes to a few hundred bytes
- ♦ Typically, from 0.3 kbps to 50 kbps
- ♦ Unsuitable for high-bandwidth applications
 - Like video, audio, or large file transfers

▷ Duty cycle restrictions

- ♦ In many regions (such as Europe), LoRa devices must obey duty cycle limits
 - They can only transmit for a small fraction of time
 - e.g., 1% of each hour
- ♦ This limits how frequently a device can send data
 - But helps prevent radio channel congestion

LoRa/LoRaWAN technical features

▷ Limited payload size

- ♦ Maximum message size in LoRaWAN is typically around 51 to 242 bytes
 - Depends on the region and LoRa data rate
- ♦ Larger messages must be split
 - Which increases latency and power consumption

▷ Network Scalability Issues

- ♦ Packet collisions increase with the density of devices
 - Because many devices may transmit simultaneously
 - Lower network reliability and throughput in denser deployments

LoRa/LoRaWAN technical features

- ▷ Latency and acknowledgement delays
 - ◆ Latency is relatively high
 - ◆ Real-time or time-sensitive applications (e.g., alarms or control systems) may experience delays
- ▷ Limited Mobility Support
 - ◆ LoRa networks are not designed for fast-moving devices
 - Like vehicles
 - ◆ The handover between gateways is not seamless, so communication can drop when devices move quickly

LoRa/LoRaWAN technical features

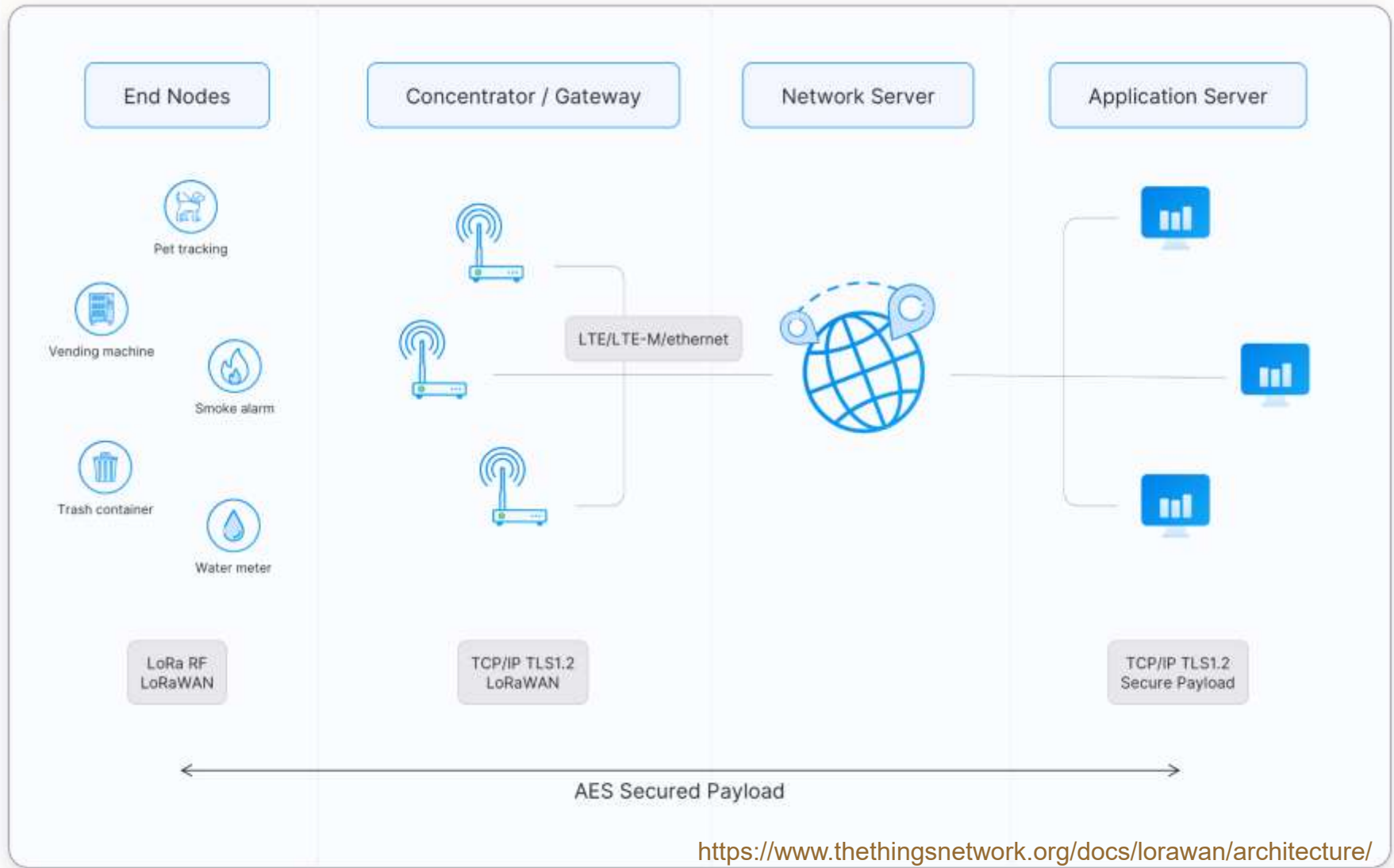
▷ Interference and range variability

- ♦ Actual range depends heavily on environment, obstacles, and antenna quality
- ♦ Urban areas may reduce the effective range to a few kilometers or less
 - Despite the “long range” name

LoRaWAN architecture

- ▷ There is not a single, closed architecture
- ▷ Independent LoRaWAN networks may coexist
 - ♦ Can be independently managed

Typical LoRaWAN network



LoRaWAN typical network elements

▷ End Devices

- ◆ Sensors or actuators
- ◆ Send LoRa messages to the gateways
- ◆ Receive messages from the gateways

▷ Gateways

- ◆ Receive messages from end devices and forward them to a Network Server

▷ Network Server

- ◆ Software running on a server that manages the entire network

▷ Application Servers

- ◆ Applications that securely processes devices' data

LoRaWAN Join Server

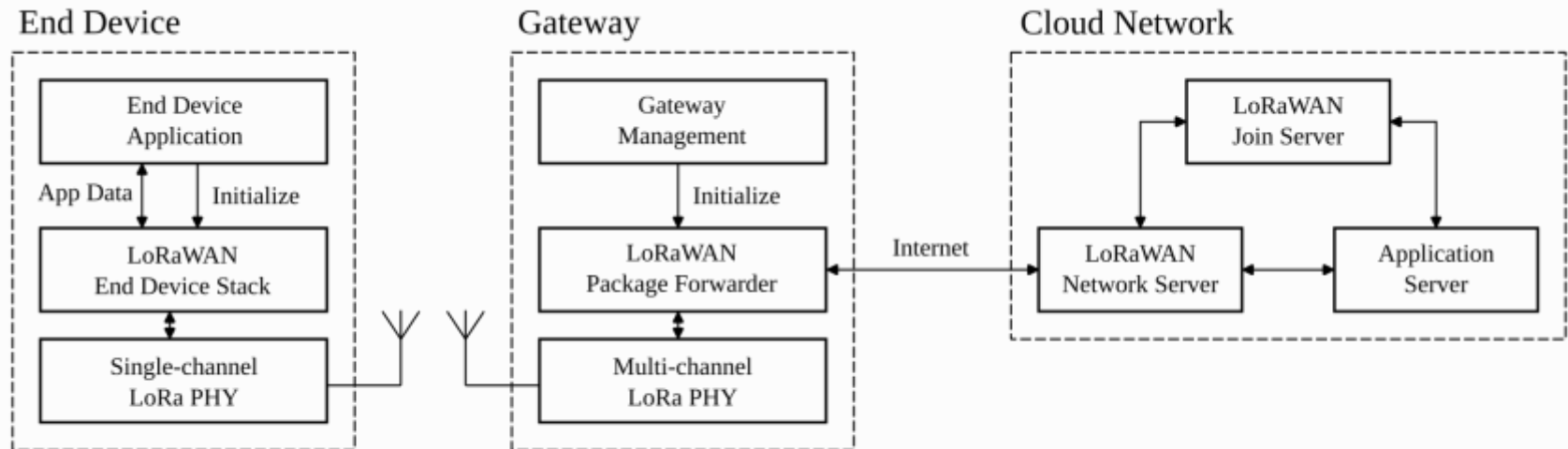
▷ Network security manager

- ◆ Maintains long-term keys that govern the network
 - Symmetric keys
- ◆ NwkKey
 - Key that bounds an End device to a Network Server
- ◆ AppKey
 - Key that bounds an End device to an Application Server

▷ Key distribution server

- ◆ Generates the confidentiality and integrity keys to be used in network sessions

LoRaWAN Join Server



<https://en.wikipedia.org/wiki/LoRa#/media/File:LoRaWAN-Architecture.svg>

LoRaWAN 1.1 message types

▷ Over-The-Air-Activation (OTAA):

- ◆ Join-request (not encrypted)
- ◆ Join-accept (encrypted)
- ◆ Rejoin-request (encrypted)

▷ Data message

- ◆ MAC commands + application data (encrypted)
- ◆ Confirmed and unconfirmed

LoRaWAN 1.1 MAC commands

- ▷ Control messages exchanged between the End Device and the Network Server
 - ♦ They can be carried in header options or payload
- ▷ They are used to manage, configure, and control how the device behaves in the network
 - ♦ e.g. adjusting data rates, confirming connectivity, or requesting status information

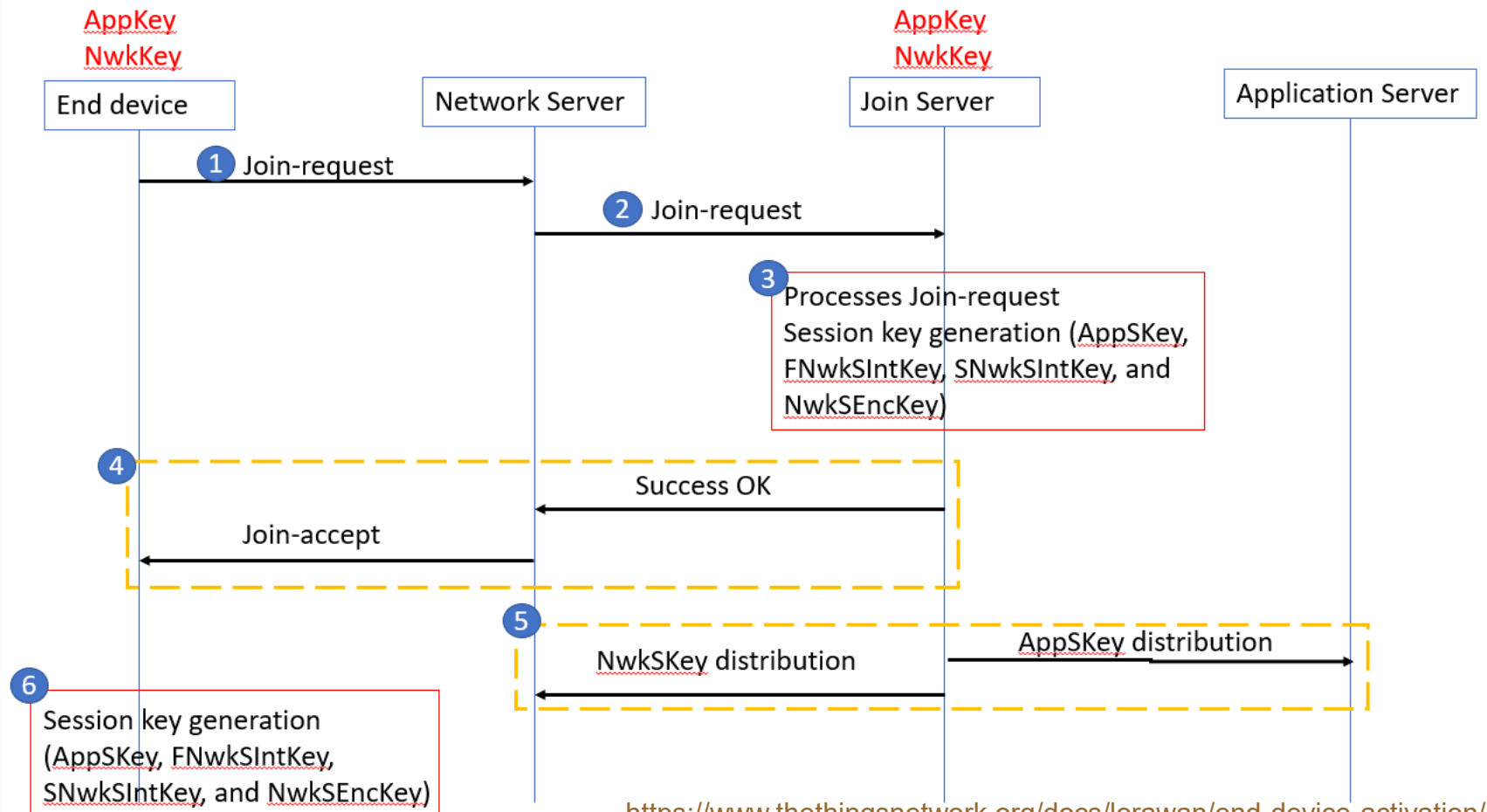
Relevant identifiers

- ▷ AppEUI
 - ◆ 64-bit globally unique application identifier
 - ◆ Uniquely identifies the entity able to process the Join Request frame
- ▷ DevEUI
 - ◆ 64-bit globally unique device identifier
- ▷ DevAddr
 - ◆ 32-bit device address assigned by the Network Server to identify the End device within the current network
- ▷ NetID
 - ◆ Identifier of a network operator
- ▷ DevNonce
 - ◆ Unique, random 2-byte value generated by the End device
 - ◆ The Network Server rejects a Join-request with a repeated DevNonce for the same DevEUI

Communication security

- ▷ Based on AES-128
- ▷ End devices and the network use a set of dynamic keys
 - ◆ These are session-based
 - ◆ Sessions are initiated when devices join the network
- ▷ Network Session Key (**NwkSKey**)
 - ◆ Shared between a device and the Network Server
 - ◆ Used for integrity control (with AES-CMAC)
 - ◆ Uplink and downlink frame counters are used to prevent message replays
 - ◆ In 1.1 this key is derived to 3 other ones for different types of messages
 - **FNwkSIntKey**, **SNwkSIntKey**, **NwkSEncKey**
- ▷ Application Session keys (**AppSKey**)
 - ◆ Shared between a device and an Application server
 - ◆ Used for payload confidentiality

Security: key derivation and distribution (version 1.1)



<https://www.thethingsnetwork.org/docs/lorawan/end-device-activation/>